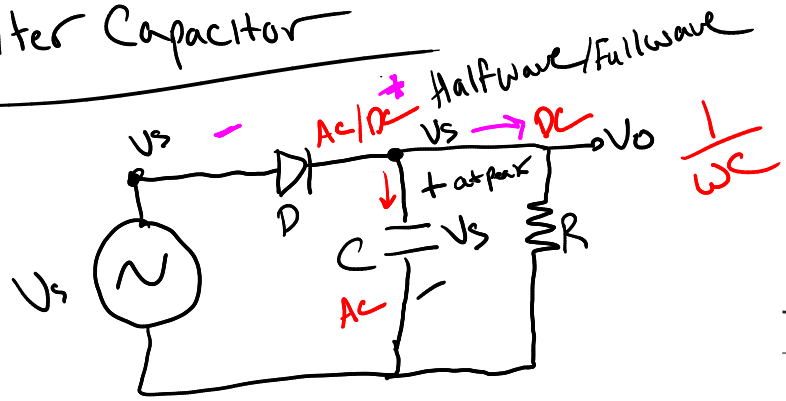
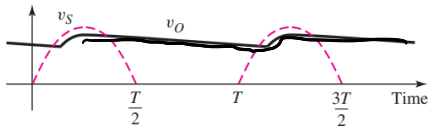
Filter Capacitor

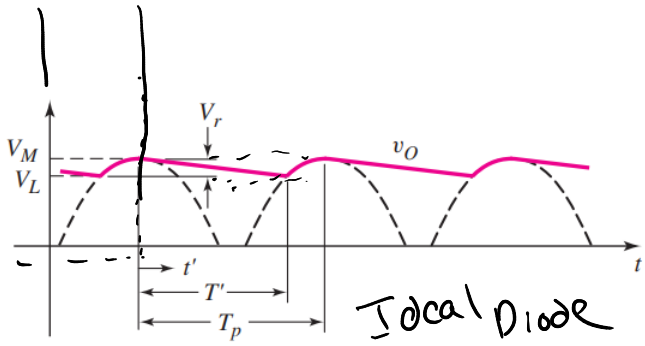


Halfwave/Fullwave



(e)

CUDZ



Ideal Diode

Figure 2.9 Output voltage of a full-wave rectifier with an *RC* filter showing the ripple voltage

* Capacitor Charges to its peak voltage
* As Vs decreases, the diode becomes reverse biased and C discharges through R

Ripple Approximation

$$V_o = V_m e^{-t'/\tau} = V_m e^{-t'/RC}$$

t': time after output has reached peak value

* The smallest output voltage is

$$V_L = V_m e^{-T'/RC}$$

where T' is the discharge time

$$V_r = V_m - V_L = V_m\left(1 - e^{-T'/RC}\right)$$

* Normally, we will want the discharge time to be

$$T' \ll RC$$

we can expand the exponential series and approximate

$$e^{-T'/RC} \approx 1 - \frac{T'}{RC}$$

Rewriting V_r

$$V_r \approx V_m\left(\frac{T'}{RC}\right)$$

This equation is very difficult to solve.
T' depends on RC

If we assume the ripple effect is small, then

T' is approximated as T_p.

T_p: Time between peak output voltage values

$$V_r \approx V_M\left(\frac{T_p}{RC}\right)$$

*For a full wave rectifier, T_p is $\frac{1}{2}$ the signal period. Therefore we can relate T_p to signal frequency

$$f = \frac{1}{2T_p}$$

// For half-wave rectifiers, T_p corresponds to the full period

Finally!! :)

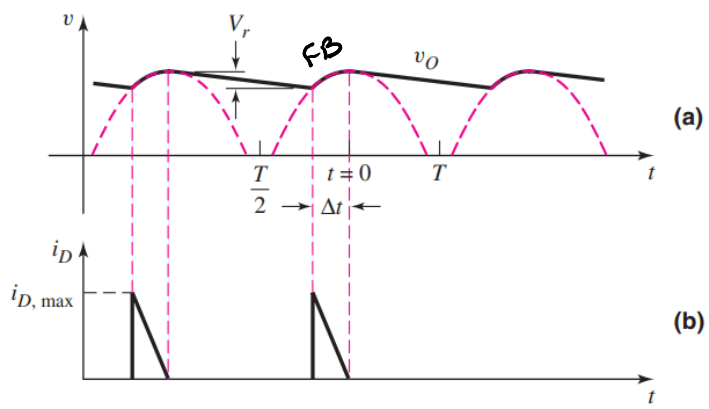
$$\boxed{V_r \approx \frac{V_M}{2fRC}}$$

So $\boxed{V_r = \frac{V_M}{fRC}}$

Example

Consider a full-wave rectifier circuit with a 60 Hz input signal and peak output voltage of $V_M = 10V$. Assume the load resistance is $R = 10k\Omega$ and the ripple voltage MUST be limited to $V_r = 0.2V$. What capacitor value is needed???

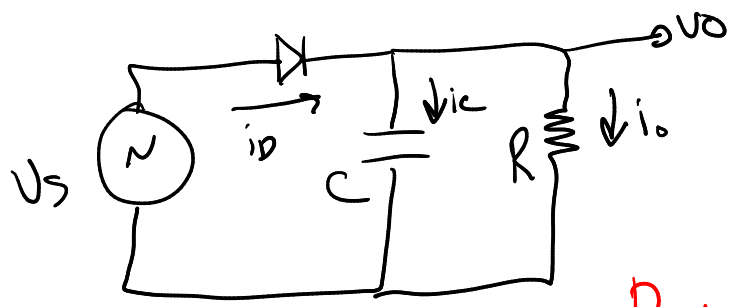
$$\Rightarrow C = \frac{V_M}{2fRV_r} = \frac{10}{2(60)(10\times10^3)(0.2)} = \boxed{41.7\mu F}$$



Ideal diodes assumed

$V_\gamma = 0$

Figure 2.10 Output of a full-wave rectifier with an *RC* filter: (a) diode conduction time and (b) diode current



Equivalent Circuit of full wave rectifier during Capacitor Charging cycle

Derivation is in the text book

$i_D = i_C + i_R$

$\Rightarrow C\frac{dv_o}{dt} + \underbrace{iR}_{v_o}$ During Diode Conduction,

$$\boxed{i_{D_{peak}} \approx \frac{V_M}{R}\left(1 + \pi\sqrt{\frac{2V_M}{V_r}}\right)}$$

* The average current through the diode during the entire cycle for a full wave rectifier is

$$\boxed{i_{D_{ave}} = \frac{1}{\pi}\sqrt{\frac{2V_r}{V_M}}\,\frac{V_M}{R}\left(1 + \frac{\pi}{2}\sqrt{\frac{2V_M}{V_r}}\right)}$$

Example

Design a full wave bridge rectifier to produce a peak output voltage of 12V, deliver 120mA to the load and produce an output with a ripple of no more than 5%. An input line voltage of 120 V_{rms} @ 60Hz is given.

Solution

The effective load resistance $R = \frac{V_o}{i_L} = \frac{12}{0.12} = 100\Omega$

Assume CVD7, the peak Vs is

$$V_{s_{max}} = V_{o_{max}} + 2V_\gamma = 12 + 2(0.7) = 13.4\,V_p$$

For a sine wave,

$$V_{s\,rms} = \frac{13.4}{\sqrt{2}} = 9.48\ V_{rms}$$

Transformer Turns ratio, $\frac{N_1}{N_2} = \frac{V_1}{V_2} = \frac{120}{9.48} = 12.7$

For a 5% ripple

$$V_r = (0.05)V_m = 0.6V$$

The required filter capacitor

$$C = \frac{V_m}{2fRV_r} = \frac{12}{2(60)(100)(0.6)} = 1667\mu F$$

The peak diode current

$$i_{Ppeak} = \frac{12}{100}\left[1 + \pi\sqrt{\frac{2(12)}{0.6}}\right] = 2.50A$$

$$i_{Dave} = \underline{\text{Math}} = 132mA$$ See equation above

$$PIV = V_{Smax} - V_o = 13.4 - 0.7 = 12.7V$$